MASTERS OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA115

MARKETING MANAGEMENT

Unit – 3

Supply and Market Equilibrium

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1. INTRODUCTION

In the last two units, we obtained an understanding of demand, the factors that influence demand (demand determinants) and the responsiveness of demand to changes in its determinants and the techniques to forecast demand. Demand for goods and/or services would be met by supply by business firms. In this unit, we will study supply and market equilibrium.

Now, we shall understand supply and its interaction with demand in achieving market equilibrium. The supply analysis is related to the behavior of producers or manufacturers. Supply of a product basically depends on cost of production and the management decision. Hence, it covers such problems as where to sell, when to sell, to whom to sell, how much to sell, at what price to sell, etc.

Demand and supply are two important concepts in economics, the knowledge of which is essential to a manufacturing firm for taking numerous decisions almost every day. These two concepts link the market behavior of consumers, producers and sellers with that of price.

1.1 Learning Objectives

After studying this unit, you should be able to:



- Describe the concept of supply and its determinants
- Interpret how demand and supply determine the market equilibrium
- Compare the different stages of market equilibrium
- Apply the knowledge of equilibrium in understanding pricing in the real world

2. SUPPLY & LAW OF SUPPLY

In this section, we will discuss the meaning of supply and the law of supply. Supply of a commodity refers to the quantities of a commodity which producers or sellers are willing to produce and offer for sale at various prices during a particular period of time. Demand and Supply are the two forces that determine the price of a commodity in the market. Supply means the quantity of goods offered for sale at a given price.

Three important aspects of supply:

1. Supply is a desired quantity I.e. how much producers are willing to sell and not the quantity they could actually sell.
2. Supply is always expressed with reference to some price.
3. Supply is a flow variable just like demand; Supply refers to an amount which the producers or sellers are willing to sell during a specific period of time.

Supply can be equal, more, or less than the current production depending upon the nature of the commodity, price and the requirements of the producers. Supply is also different from stock. Stock is the total volume of a commodity which can be brought into the market for sale at short notice and supply means the quantity which is actually brought into the market. For perishable commodities, like fish and fruits, supply and stock are the same because they cannot be stored. The commodities which are not perishable can be held back if prices are not favorable and released in large quantities when prices are favorable. In short, stock is a potential supply.

Definitions of supply:

According to Thomas, "The supply of goods is the quantity offered for sale in a given market at a given time at various prices".

According to Prof. McConnell – "supply may be defined as a schedule which shows the various amounts of a product, which a producer is willing to and able to produce and make available for sale in the market at each specific price in a set of possible prices, during some given period."

2.1. Individual and Market supply:

Individual Supply: Quantity of a commodity which one producer is willing to produce and offer for sale. It refers to the quantity of a commodity which a firm is willing to produce and offer for sale at a particular price during a specified period.

Market supply: The quantity that all the producers are willing to produce and offer for sale at a particular price during a specified period is known as market supply or industry's supply.

2.2. Determinants of supply

Apart from price, many other factors bring about changes in supply. Among them, the important factors are:

1. ***Natural factors* –** Favorable natural factors like good climatic conditions and timely, adequate, well distributed rainfall results in higher production and expansion in supply. On the other hand, adverse factors like bad weather conditions, earthquakes, pests, droughts and untimely, ill-distributed, inadequate rainfall, etc., may cause decline in production and contraction in supply.
2. ***Change in techniques of production* –** An improvement in techniques of production and use of modern, highly sophisticated machines and equipment will go a long way in raising the output and expansion in supply. On the contrary, primitive techniques are responsible for lower output and hence lower supply.
3. ***Cost of production* –** Given the market price of a product, if the cost of production rises due to higher wages, interest and price of inputs, supply decreases. If the cost of production falls on account of lower wages, interest and price of inputs, supply rises.
4. ***Prices of related goods* –** If prices of related goods or substitutes fall, the seller of a given commodity offers more units in the market even though the price of his product has not gone up. The opposite will be the case when the price of related goods rises. Similarly, a producer produces A goods and if the market price of B goods increases, then the producer increases the production of B goods. Therefore, the supply of B goods increases. The supply of a commodity varies inversely to a change in the price of a substitute good in production.

   In case of Complementary goods, especially for joint demand, the production and sale of one product may lead to production and sale of other products also.

5. ***Government policy –*** When the government follows a positive policy, it encourages production in the private sector. Consequently, supply expands. For example, granting subsidies, development rebates, tax concession, etc. On the other hand, output and supply cripples when the government adopts a negative policy, for example, withdrawal of all concessions and incentives, imposition of high taxes, introduction of controls and quota system, etc. Similarly, Indirect taxes such as sales tax, excise duty, VAT are likely to increase prices of commodities and which in turn increase the cost of production therefore will lead to a decrease in supply. Subsidies to firms encourage the producers to produce certain goods as this leads to a decrease in the cost of production, thereby increasing the supply.
6. ***Monopoly power –*** Supply tends to be low when the market is controlled by monopolists, or a few sellers as in the case of oligopoly. Generally, supply would be more admis competitive conditions.
7. ***Number of sellers or firms –*** Supply would be more when there are a large number of sellers. Similarly, production and supply tend to be more when production is organized on large scale basis. If the rate or speed of production is high, supply expands. The opposite will be the case when the number of sellers is less, with small scale production and low rate of production.
8. ***Discovery of new sources of inputs –*** Discovery of new sources of inputs helps the producers to supply more at the same price and vice- versa.
9. ***Improvements in transport and communication –*** This will facilitate free and quick movements of goods and services from production centers to marketing centers. This leads to an expansion in the market size as it motivates the producers to increase production and supply.
10. ***Expectation of Future rise in prices –*** When sellers anticipate a further rise in price, the current supply tends to fall. The opposite will be the case when the seller expects a fall in price. Speed of production is high, supply expands. The opposite will be the case when the number of sellers is less, with small scale production and low rate of production.
11. ***Agreement among producers:*** In the case of certain industries or product markets the producers form a pool or an association and enter into an agreement to restrict the supply of a commodity to earn large profits. They create artificial scarcity of the commodity as a result of which the supply is decreased.

2.3. Supply function

The law of supply and supply schedule explains only the direct relationship between price and supply. Mathematically, S = f (P). This relation analyses the impact of a change in price on quantity supplied. Supply of a product also depends upon many factors apart from a change in prices. When we analyze the influence of these factors on supply, the supply schedule will be converted into a supply function.

The supply function is a comprehensive one as it analyses the causes of changes in supply in a detailed manner. Mathematically, a supply function can be represented in the following manner.

Sn = f (Pn, P1...Pn-1, Gf, Fi...Fm,T,E, Gt,N, M....)

Here,

Sn = Quantity supplied of a commodity

f= is a symbol showing the relationship between supply of a commodity n and all its determinants

Pn = is the price of the commodity

Pn-1 = are the prices of the commodities other than n

Gf = is the goal of the firms

Fi...Fm = are the prices of different factors of production

T =is the technique of production

E =is the symbol for expectation about future prices.

Gt =is the taxation policy of the government

N= is the expression of the natural factors

M =stands for means of transportation.

2.4. The Law of supply

The law of supply or supply hypothesis states that, other things remaining the same, the quantity of any commodity that firms will produce and offer for sale rises with a rise in its price and falls with a fall in its price.

Normally, a seller supplies more units of a commodity at a higher price and vice versa. Given the cost of production, profits are likely to be high at higher prices. The higher the price, the greater the inducement is to the producers to produce and sell more and make more profits. Hence, more is supplied at higher prices and less is supplied at lower prices. This relationship between the price and the quantity supplied is popularly known as the law of supply. It states that "Other things remaining constant, the quantity supplied varies directly with the price i.e., when the price falls, supply will contract and when price rises, supply will extend". According to S.E.Thomas, "a rise in price tends to increase supply and a fall in price tends to reduce it." There is a functional relationship between supply and price. Mathematically S = F (P). The law of supply is based on a number of assumptions.

The other things are nothing but the determinants of supply which should remain constant for the law to operate are as follows:

1. Number of firms, the scale of production and the speed of production
2. Availability of other inputs
3. Techniques of production
4. Cost of production
5. Market prices of other related goods
6. Climate and weather conditions
7. Taxation policy
8. Goals of production

Thus, in this section, we have discussed the meaning and the law of supply.

2.5. Supply schedule

Supply schedule is a tabular representation of different quantities of a commodity supplied at varying prices. It represents the functional relationship between quantity supplied and price. It is strictly prepared with reference to the price of a given commodity. This statement shows various quantities which producers are willing to produce and sell at various alternative prices during a given period of time. The supply schedule is of two types.

1. Individual supply Schedule
2. Market supply schedule

1. Individual supply schedule: is a table which shows various quantities of a commodity that an individual producer or a firm would offer for sale at different prices during a given period.

   The imaginary supply schedule given in table 4.1 shows that as price rises, supply extends and as price falls, supply contracts. For instance, 0.75 paisa is the minimum price to be charged per unit, because it equals the cost of production. No producer would like to charge the cost price to customers. Hence, supply is zero at this price. It is called reserve price.

The table below shows the units supplied by an individual producer or a firm.

Table 4.1: Supply Schedule

Price in Rs.	Quantity supplied in Units
5-00	500
4-00	400
3-00	300
2-00	200
1-00	100
0-75	00

Market supply schedule

The total quantity of a commodity supplied at different prices in a market by the whole body of sellers is called a market supply schedule. It refers to the aggregate behavior of the market rather than mere totaling of all individual supply schedules. It is the horizontal summation of the individual supply schedule. Table 4.2 shows the market supply schedule.

Table 4.2: Market Supply Schedule

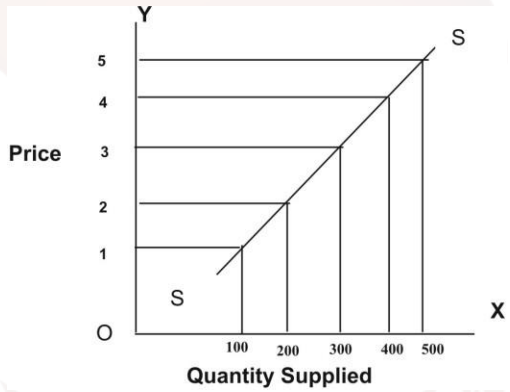
Price in Rs.	Quantity Supplied in Units			Total (A+B+C)
	A	B	C	
5.00	500	600	700	1800
4.00	400	500	600	1500
3.00	300	400	500	1200
2.00	200	300	400	900
1.00	100	200	300	600

The market supply schedule helps a firm to formulate its sales policy by manipulating the prices. It helps the management to know how much sales can be increased by raising the price without losing the demand for the product.

2.6. Supply curve

The supply curve is a geometrical representation of the supply schedule. The upward sloping curve clearly indicates that as price rises, quantity supplied expands and vice versa. Figure 4.1 depicts the supply curve.

Figure 4.1: Supply Curve



Price of Good X (Rs per Kg)	Quantity supplied of Good X (Kg per month)	Points on supply curve
10	5000	A
15	10,000	B
20	16000	C
25	25000	D
30	35000	E

Meaning of Supply Curve:

A supply curve is a diagrammatic presentation of the law of supply. The information in a supply schedule is projected on a supply curve. There are two types of supply curves:

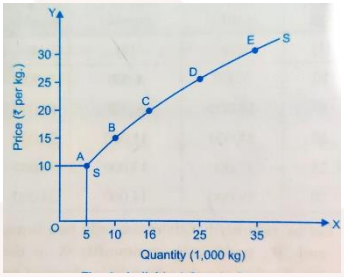
1. Individual Supply Curve
2. Market Supply curve

Individual Supply curve: An individual supply curve is defined as the curve which shows various quantities of a given commodity which an individual producer is willing to supply at different prices

during a given period of time assuming no change in the factors other than its own price affecting supply of the commodity.

Individual supply schedule

Source: Economics from Dk Sethi



Individual supply curve

Source: Economics from Dk Sethi

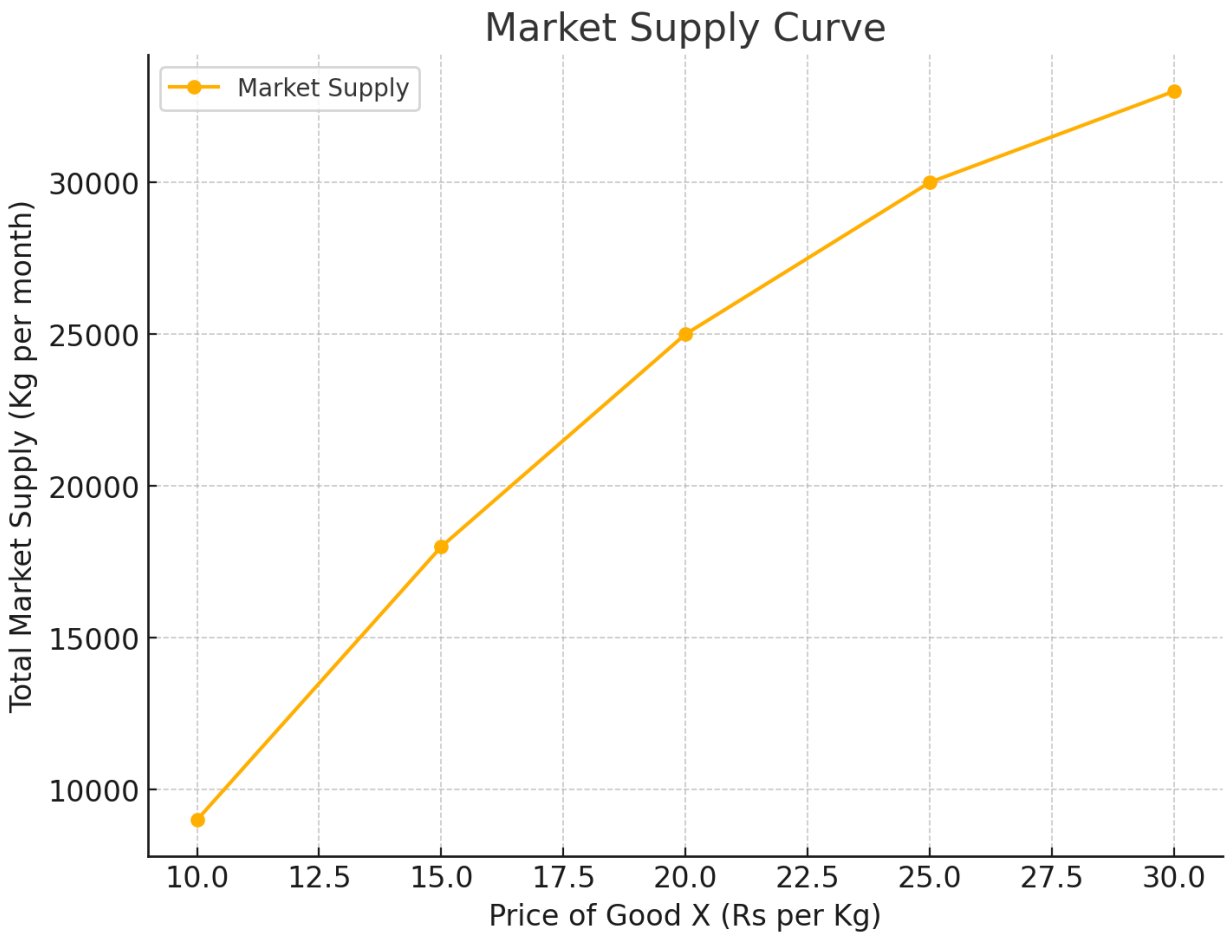
The individual supply curve shows five points A,B,C,D,and E corresponding to each price-quantity combination. Where we show price, on Y-axis and quantity supplied on X axis. By joining points A,B,C,D,and E we arrive at a curve SS. The supply curve is a positive sloping supply curve SS that shows a direct relationship between the price and the quantity supplied.

2. **Market Supply Curve:** is a curve that shows various quantities of a commodity which all the producers are willing to produce and sell at different prices during a given period of time assuming factors other than the own price of the goods given.

Market supply schedule for good X

Let us assume that there are only two firms, Firm A and Firm B producing commodity X in the market. The market supply schedule is obtained by adding the quantity supplied by Firms A and B at different prices. For instance, 5000 + 4000 = 9000. We plot column 1 and column 4 on the graph.

Price of Good X (Rs per Kg)	**Quantity supplied by Firm A (Kg per month)**	**Quantity supplied by Firm B (Kg per month)**	Total Market supply (Kg per month)
10	5000	4000	9000
15	10,000	8000	18000
20	14000	11000	25000
25	17000	13000	30000
30	19000	14000	33000



Market supply curve

Here is the market supply curve based on the data provided. The graph illustrates the relationship between the price of Good X (Rs per Kg) and the total market supply (Kg per month). As the price increases, the total market supply also increases, showing a typical upward-sloping supply curve.

The market supply curve SS is a positive sloping curve showing a direct relationship between price and quantity supplied by all the producers in the market. In this case it is firm A and B. With price being represented on the Y axis and Quantity on the X axis. The supply curve SS represents the total quantity supplied by both Firm A and B as per the market supply schedule.

2.7. Reasons behind the Positive slope of the supply curve:

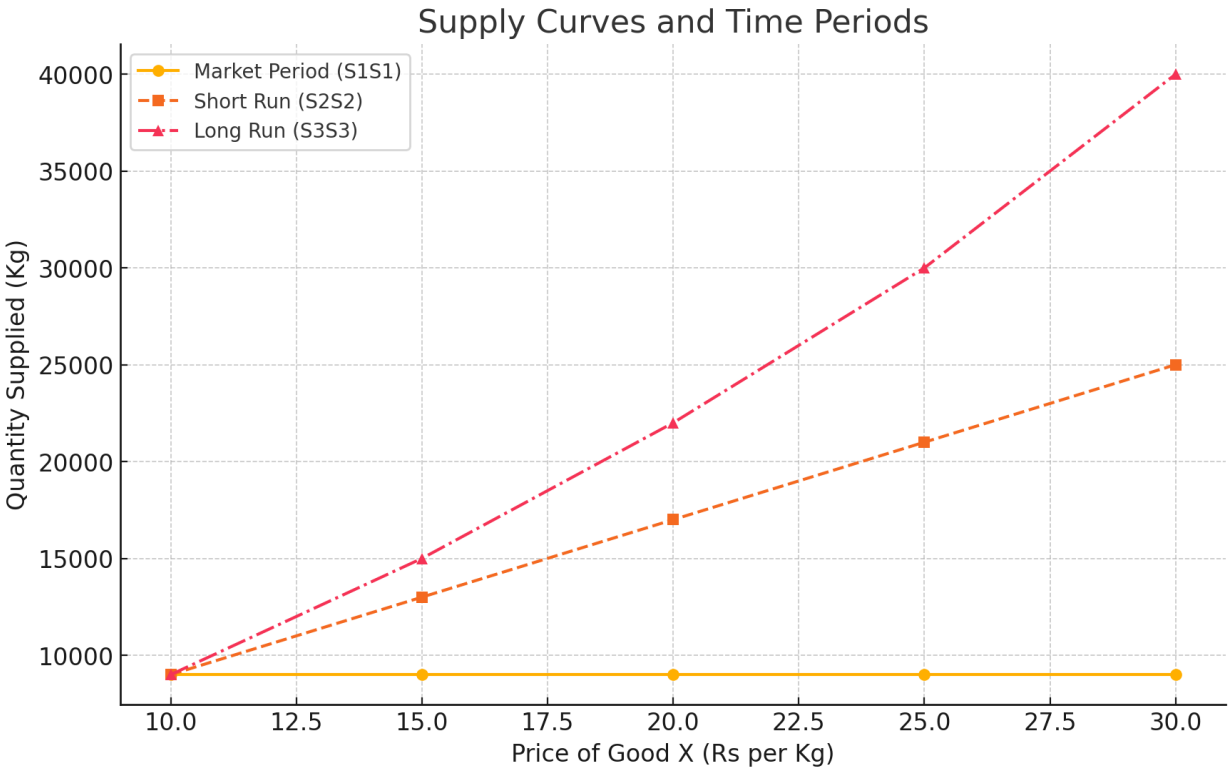
1. Other things remain constant, the higher the price the greater the profit that can be earned. This acts as an incentive to producers to produce and supply a larger quantity of the commodity. Thus, the higher the price, the greater the incentive of profit and larger the quantity supplied.

2. The marginal cost of production increases with an increase in output because of the law of diminishing marginal returns. Therefore, as production output increases the cost increases and due to this the producer is only ready to supply a larger quantity of the commodity at a high price to cover the high cost of production.
3. A rise in the price of commodity A encourages other prospective producers to produce more of commodity A. Producers who are producing other commodities may also switch to producing commodity A. There by increasing supply of the commodity A.

2.8. Time period and supply:

Market (Very short) period: Market period is a very short period in which the supply is fixed. The maximum supply during this period is the stock of the goods which have already been produced. This is in the case of perishable goods where firms cannot adjust their output to any change in price. For instance, the stock of vegetables grown in a lot is supplied to the vegetable market, there cannot be any change in the supply nor can their sale be postponed as the vegetables can perish. **Short-Run:** is the period in which the supply can be adjusted to some extent. Producers can increase production by running the existing plant for longer hours or two shifts instead of one shift. A short run period is not long enough for a new plant to be set.

Long Run: is the period in which firms can build new plants, set up or close down an old plant or enter a new industry by setting up new plants and machinery. It is possible to bring in a large change in quantity supplied in the long run.

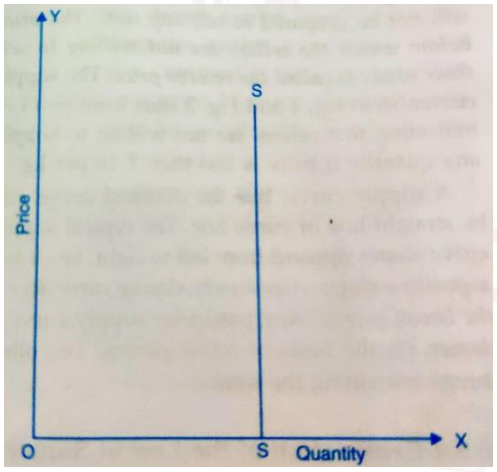
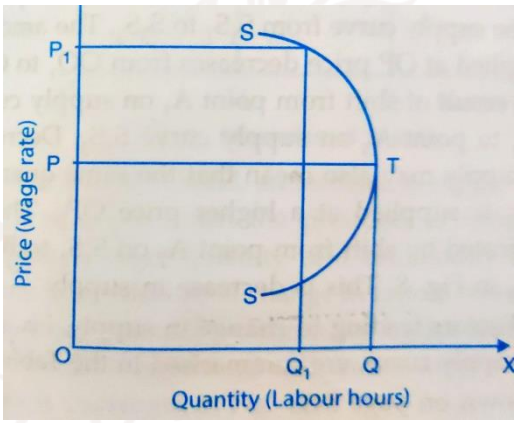


Supply curves and Time period graph

3. EXCEPTIONS TO THE LAW OF SUPPLY

In this section, we will discuss the exceptions to the law of supply. Generally, supply expands with the rise in price and contracts with the fall in price. But under certain exceptional circumstances, a violation of the supply curve is observed.

There are two exceptional shapes of the supply curve:

1. Vertical or fixed supply curve	**2.Backward sloping Supply curve**
In certain commodities the quantity supplied remains fixed. For e.g., Land, Paintings, Antiques, Stamps. Or the supply is fixed in the short run and can be increased in the long run For e.g., Agricultural crops, the wheat supply is fixed but can be increased in the long run.	This means that a smaller quantity is offered when the price is high and a larger quantity is offered when the price is low. E.g., In labor supply, If the wage rate increases the worker prefers to work for more hours initially to earn more income. But as we continue if the wage rate reaches a higher point the worker prefers to work fewer hours and prefers leisure or rest as he would have earned enough by then.
 **Vertical Supply Curve**	 **Backward Supply Curve**
In such a case SS the supply curve is parallel to OY the Price.	When the price is OP, the supply is OQ but when the price rises to OP1 then the supply falls to OQ1. SS curve shows a negative slope beyond point T.

Source: Economics by DK Sethi

in spite of rise in price, supply may not expand or at a lower price, more quantity may be sold. In this case, the supply curve slopes backward. Figure 4.2 depicts the change in supply with respect to changes in price.

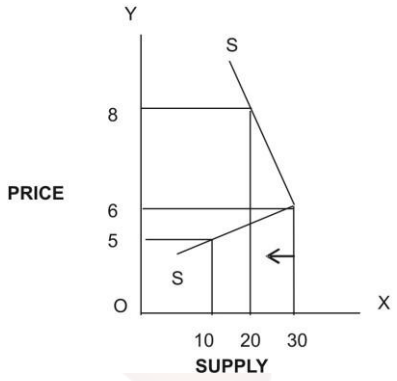


Figure 4.2: Supply Curve – Changes in Price

In figure 4.2, when price is Rs. 5.00, 10 units are sold and when the price is Rs. 6.00, 30 units are sold. But when price rises to Rs. 8.00 quantity supplied falls from 30 units to 20 units.

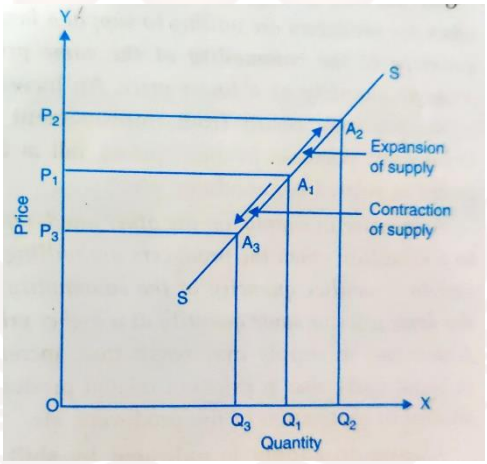
Some of the exceptions to the law of supply are as follows:

1. If the seller is badly in need of money, he will sell more even at lower prices.
2. If the seller wants to get rid of his products, he will then sell more at reduced rates.
3. When a further heavy fall in price is anticipated, the seller may and sell more at a current lower price.
4. In the case of an auction, the auctioneer is not interested in maximizing profits by selling more units at a higher price. Here, the price is determined by the bidder while selling an item in an auction, the auctioneer may have some other motives to sell the product. Thus, an auction sale is an exception to the law of supply.

4. CHANGES OR SHIFTS IN SUPPLY – EXPANSION AND CONTRACTION

In this section, we will discuss the changes and shifts in supply. When supply of a product changes only due to a change in the price of that product alone, it is called either expansion or contraction in supply. **Expansion** in supply means more quantity is supplied at a higher price. **Contraction** in supply means, less quantity is supplied at a lower price.

This tendency can be represented through a single supply curve. In this case, the seller will be moving either in the upward or downward direction along with the same supply curve. The expansion and contraction of supply graph depicts the tendency of shifts in supply through a single supply curve.



Expansion and Contraction of supply graph

Source: Economics by DK Sethi

The expansion and contraction of supply graph shows that when Price is OP1 the quantity supplied is OQ1.Now if the price rises to OP2, the quantity supplied rises to OQ2. The movement from point A1 to A2 in the upward direction on the given supply curve SS represents “Expansion of supply”

If the price of a commodity falls from OP1 to OP3 then the quantity supplied also falls to OQ3. The movement from A1 to A3 in the downward direction on the given supply curve SS is “Contraction in supply”.

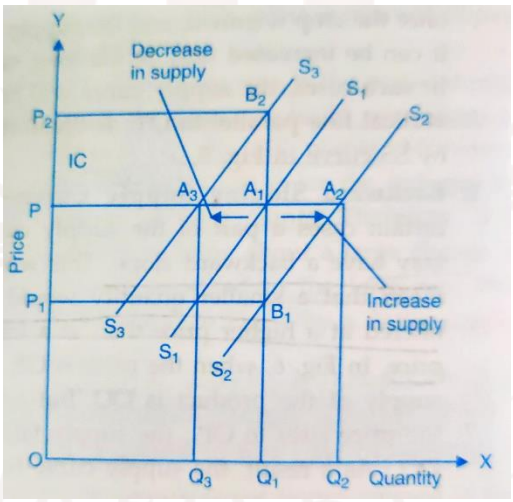
5. CHANGES OR SHIFTS IN SUPPLY – INCREASE AND DECREASE

Supply of a product may change due to changes in other factors. If supply changes not because of changes in price, but because of changes in other determinants, then, it will be a case of either increase or decrease in supply.

Increase in supply

It implies more supply at the same price or the same quantity of supply at a lower price. In this case, we have to draw a new supply curve.

When producers are willing to supply a larger quantity of the commodity at the same price or same quantity at a lower price. An increase in supply may be due to improvement in technology, or a decrease in the cost of production.



Shift in supply - Increase and decrease in supply

Source: Economics DK Sethi

A right ward shift of supply curve from S1S1 to S2S2 indicates increase in supply.

a. At price OP the producers shift from point A1 on S1S1 curve to point A2 on S2S2 as they have increased the quantity supplied.

   The producers offer a larger quantity from OQ1 to OQ2 at the same price OP

b. Increase in supply can also mean the same quantity OQ1 is supplied at a lower price OP1

This is shown by a shift from A1 on S1S1 curve to B1on S2S2 curve

A left ward shift from S1S1 to S3S3 indicates a decrease in supply.

The quantity supplied at OP price decreases from OQ1 to OQ3 as a result there is a shift from point A1 on S1S1 to A3 on S3S3 curve.

c. Decrease in supply may also mean the same quantity is offered at OQ1 at OP2 higher price.
d. This is indicated by shifting from point A1 on S1S1 to B2 on S3S3. This is a decrease in supply

Factors causing change in supply

Shift Factors	*Increase in Supply (Rightward shift of supply curve)*	*Decrease in Supply (Leftward shift of supply curve)*
1. **Technology**	1. Improvement in techniques of production.	1. Use of less efficient techniques of production.
2. **Input Prices**	2. Fall in input prices.	2. Rise in input prices.
3. **Prices of other goods**	3. Fall in the prices of other goods.	3. Rise in the prices of other goods.
4. **Taxes and subsidies**	4. Lowering of taxes on production or raising production subsidies.	4. Raising taxes on production or lowering production subsidies.
5. **Producers' price expectations**	5. Expected fall in prices of goods in future.	5. Expected rise in prices of goods in future.
6. **Number of sellers**	6. The number of sellers increases.	6. The number of sellers decreases.

Source: DK Sethi, Economics

Managerial uses of the law of supply

The law of supply helps a producer to take decisions with respect to the following points:

1. Identifying the type of product, to produce and sell. This point encapsulates answers to the following queries:
   I. What quantity to sell?
   II. At what price to sell?
   III. When to produce
   IV. Where to sell.
   V. For whom to sell or offer the product to.
2. Maintaining a balance between stock and supply.
3. Preparing the sales budget policy.

4. Estimating the present and future expected revenue and profit levels.
5. Analyzing the effects of taxes on total sales in the market.
6. Analyzing the impact of various government policies on the supply of a product.
7. Identifying the factors which affect supply of a product. Supply function is also described as shifts in supply.

INSPIRED BY LIFE

6. ELASTICITY OF SUPPLY

In this section, we will discuss the elasticity of supply. It is a parallel concept to elasticity of demand. Price Elasticity of Supply measures the degree of sensitivity or responsiveness of the quantity supplied of a commodity to a change in its price. In short, it measures the degree of adjustability of supply to a given change in price of a product. Usually, the elasticity of supply is positive.

The formula to calculate elasticity of supply is as follows:

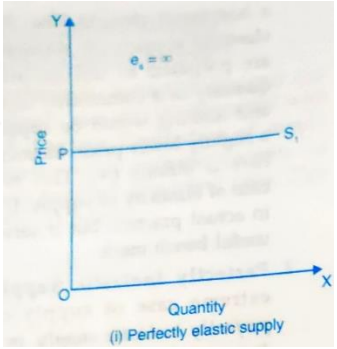
$$e_s = \frac{\%\text{ change in quantity Supplied}}{\%\text{ Change in Price}}$$

6.1. Types of elasticity of supply:

Just like elasticity of demand, elasticity of supply can be equal to infinity, zero, greater than one, lower than one and equal to one. Let us discuss the types of elasticity of supply.

Infinite or perfectly elastic supply

Supply is said to be perfectly elastic when the quantity supplied of a commodity responds by an infinite level to a very small change in price. A slight change in price leads to immeasurable changes in supply. Hence the supply curve would be a horizontal or parallel line to X axis. Graph (I) shows that sellers are ready to sell infinitely large quantities of a commodity at OP Price and nothing would be supplied at all at a slightly lower price

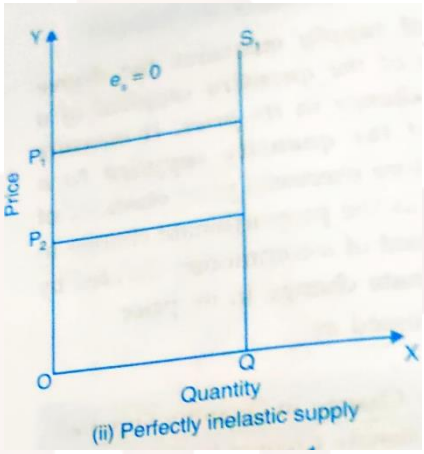


Graph (I) Perfectly elastic supply

Source: Economics DK Sethi.

Absolutely or perfectly inelastic supply

When supply of a commodity remains constant or has no responsiveness and does not change, no matter how large a price change takes place, it is said to be absolutely or perfectly inelastic supply. Here the supply curve tends to be a vertical straight line. ES = 0 (zero). Graph (ii) depicts a perfectly inelastic supply curve.

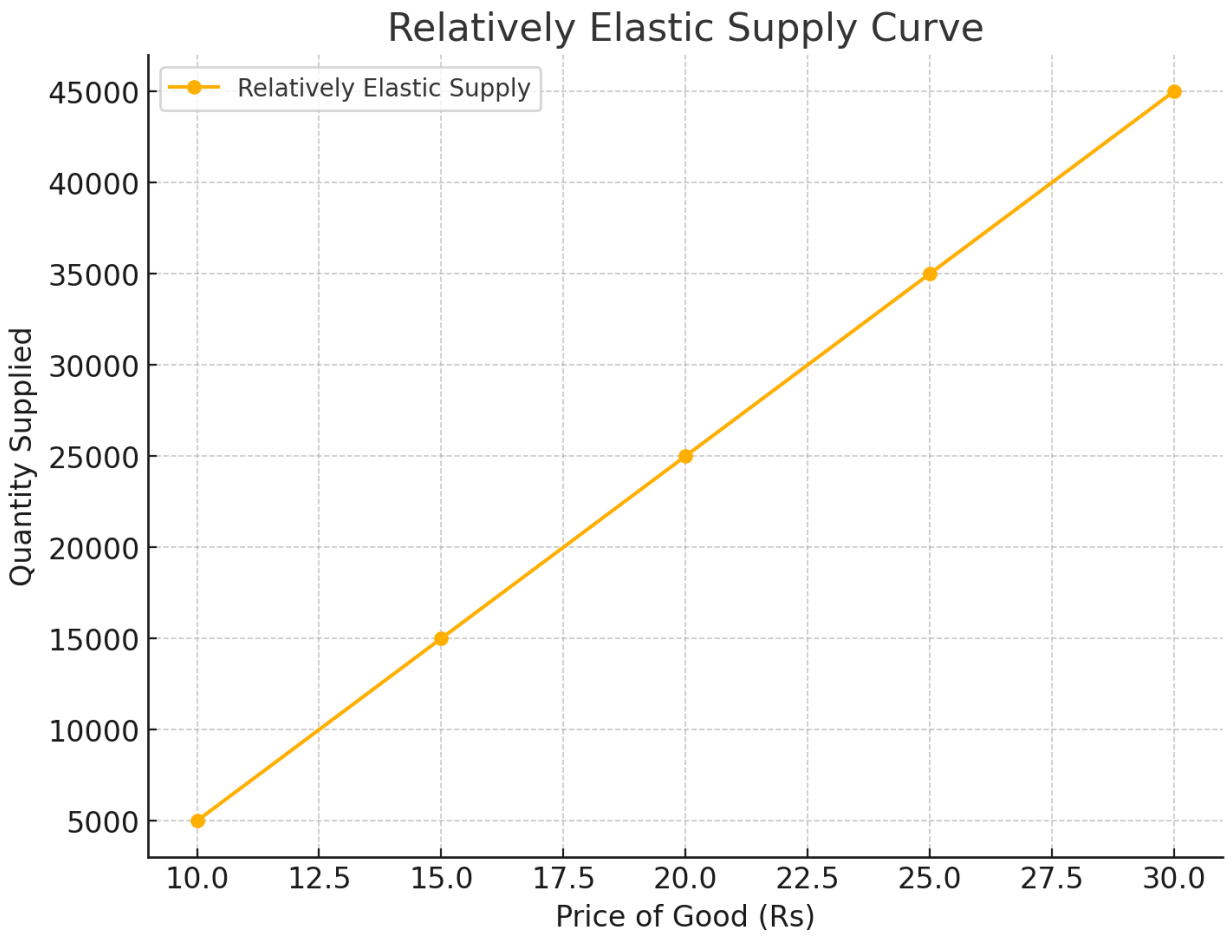


Graph (ii): Perfectly Inelastic Supply Curve

Source: Economics DK Sethi.

Elastic or relatively elastic supply

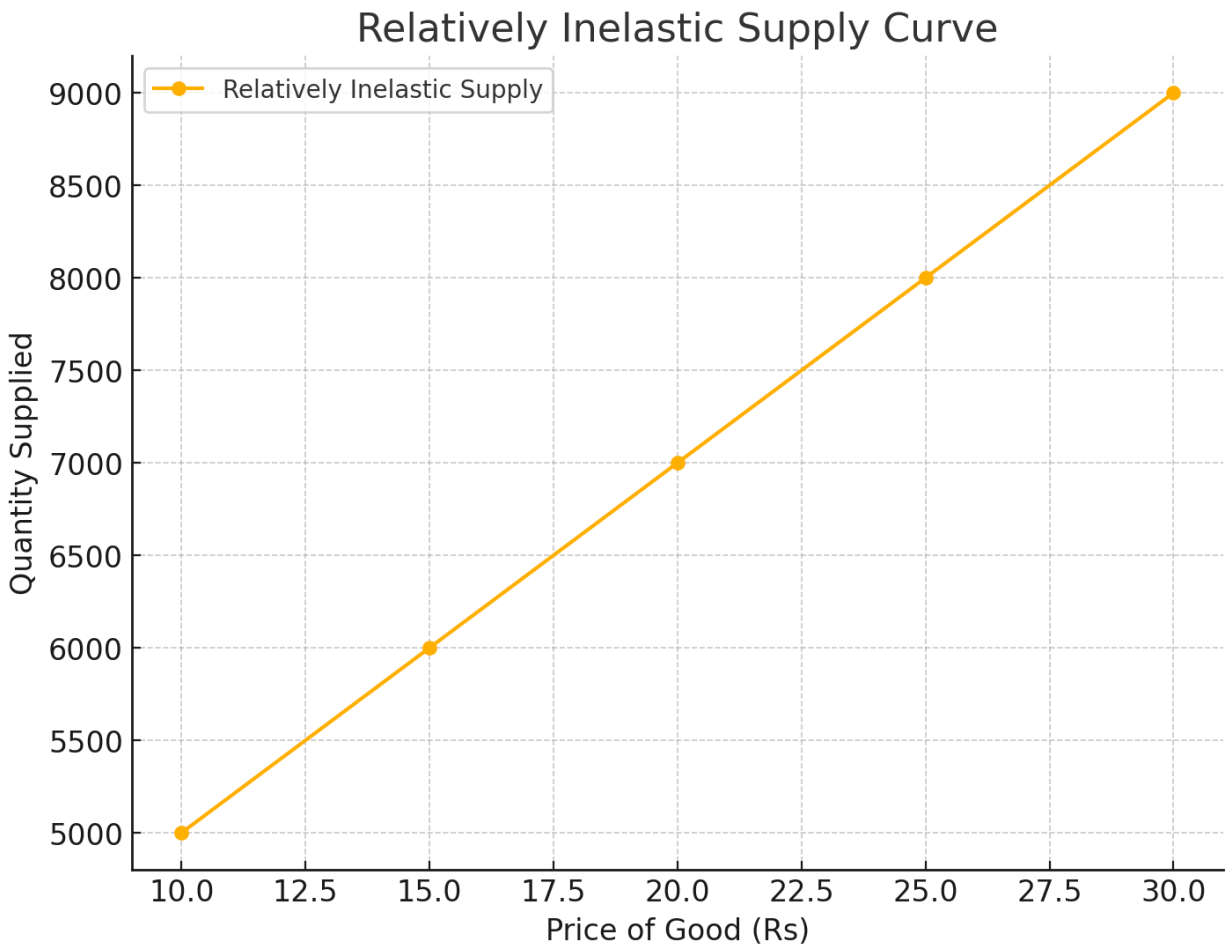
When the percentage change in the quantity supplied is more than proportional to the percentage change in price, it is called elastic supply. Here, the elasticity of supply is greater than one. In that case, the supply curve is flatter and is more inclined to the X axis.



Here is the graphical representation of a relatively elastic supply curve. The curve shows that a small change in price leads to a significant change in the quantity supplied, indicating high elasticity of supply. This is typical in industries where production can quickly adjust to price changes, such as consumer goods or services with flexible manufacturing processes.

Relatively inelastic or Inelastic supply

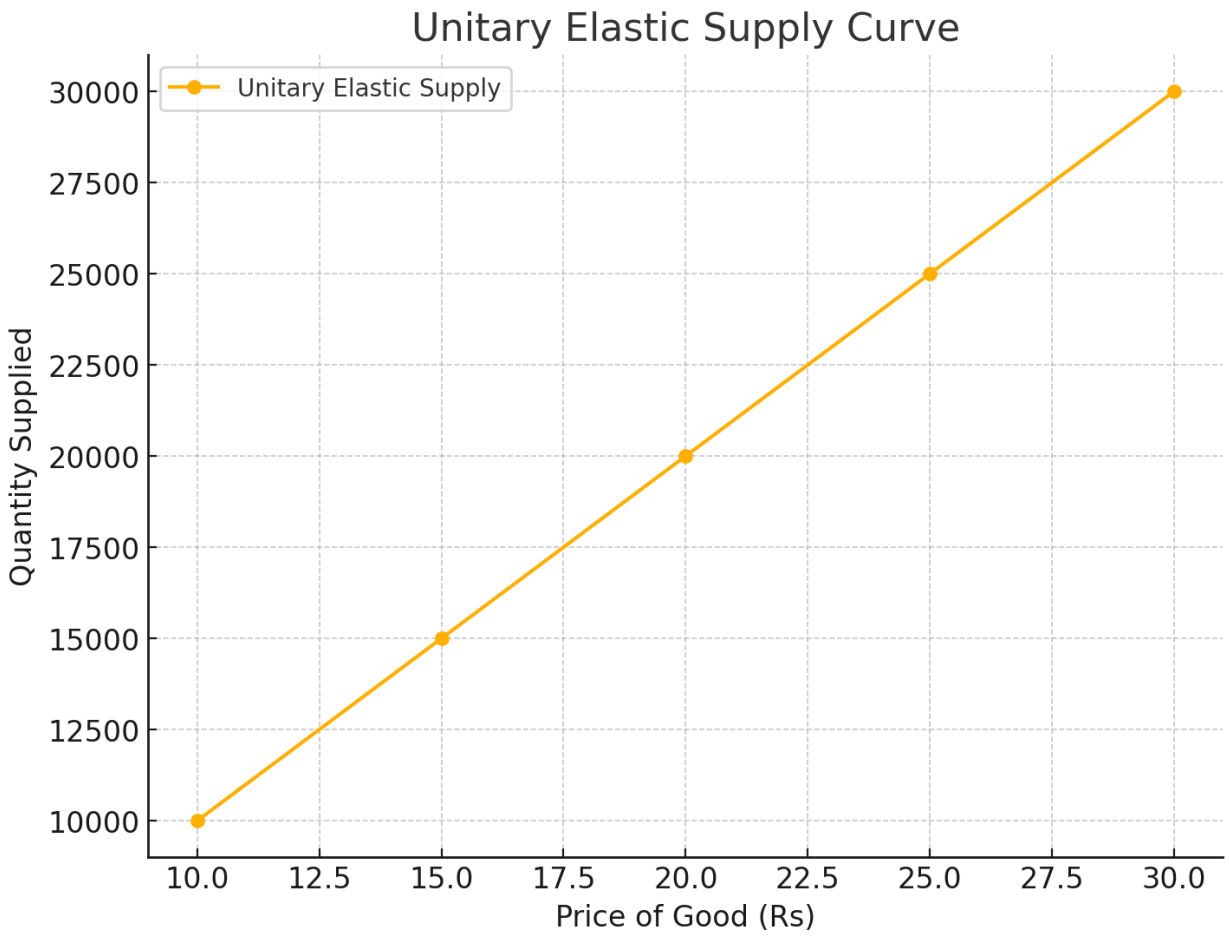
When the percentage change in quantity supplied is less than proportional to a percentage change in price, then, elasticity of supply is said to be Inelastic or relatively inelastic. Elasticity is less than one. Here, the supply curve is a steeply rising one.



Here is the graphical representation of a **relatively inelastic supply curve**. The curve shows that even with a significant change in price, the quantity supplied changes only slightly. This typically occurs in industries where production adjustments are difficult or time-consuming, such as agriculture or industries with fixed capacities in the short run.

Unitary elastic supply

If proportionate change in supply is exactly equal and proportional to the change in price, then the elasticity of supply is equal to one.



Here is the graphical representation of a **unitary elastic supply curve**. The curve demonstrates a proportional relationship between price and quantity supplied, meaning a percentage change in price results in an equivalent percentage change in supply. This behavior represents a balanced responsiveness of suppliers to price changes.

7. FACTORS DETERMINING THE ELASTICITY OF SUPPLY

In this section we will discuss the factors determining the elasticity of supply. The factors determining the elasticity of supply are as follows:

1. ***Time period*** – Time has a greater influence on elasticity of supply than on demand. Generally, supply tends to be inelastic in the short run because time available to organize and adjust supply to demand would be insufficient. Supply would be more elastic in the long run.
2. ***Availability and mobility of factors of production*** – When factors of production are available in plenty and freely mobile from one occupation to another, supply tends to be elastic and vice versa.
3. ***Technological improvements*** – Modern methods of production expand output and help reduce the cost of production. Hence, supply tends to be elastic. Old methods reduce output and supply tends to be inelastic.
4. ***Cost of production*** – If the cost of production rises rapidly as output expands, then there will not be much incentive to increase output as the extra benefit will be choked off by the increase in cost. Hence, supply tends to be inelastic and vice versa.
5. ***Kinds and nature of markets*** – If the seller sells the product in different markets, supply tends to be elastic in any one of the markets because a fall in the price in one market will induce the seller to sell in another market. Again, if the seller is producing several types of goods and can switch over easily from one to another, then each product will be elastic in supply.
6. ***Political conditions*** – Political conditions may disrupt production of a product. In that case, supply tends to become inelastic.
7. ***Number of sellers*** – Supply tends to become more elastic if there are more sellers freely selling their products and vice versa.
8. ***Prices of related goods*** – A firm can charge a higher price for its products, if prices of other products are higher and vice versa.
9. ***Goals of the firm*** – If the seller is happy with a small output, supply tends to be inelastic and vice versa.

Thus, several factors influence the elasticity of supply.

8. PRACTICAL IMPORTANCE OF ELASTICITY OF SUPPLY

In this section we will discuss the practical importance of elasticity of supply. The following points highlight the practical importance of elasticity of supply:

1. The concept of elasticity of supply is of great importance to the finance minister while formulating the taxation policy of the country. If the supply is inelastic, the imposition of tax may not bring about any change in the supply. If supply is elastic, reasonable taxes are to be levied.
2. The price of a commodity depends upon the degree of elasticity of demand and supply.
3. It is used in the theory of incidence of taxation. The money burden of taxation is shared by the tax payers and the sellers in the ratio of elasticity of supply and demand.

Thus, elasticity of supply has both theoretical as well as practical applications.

SELF ASSESSMENTS QUESTIONS – 1

Fill in the blanks:	
1	Stock is also known as _____ .
2	Supply curve generally slopes _____ .
3	_____ shows the relationship between price and quantity supplied of a particular product.
4	The supply curve shifts when there is a change in quantity supplied due to factors other than _____ .
5	When the supply increase the supply curve shifts to the_____ ..
6	Supply tends to be _____ in the short run.
7	The tabular representation of different quantities of commodities supplied at varying prices is called _____ .

True or False Questions	
i)	Supply and stock have the same effect on quantity offered for sale.
ii)	New technology can never change the quantity supplied.
iii)	Other things remaining the same, an increase in input costs leads to a decrease in the quantity supplied.
iv)	The supply curve depicts the price of the good and quantity produced at various levels of price.
v)	The supply curve slopes downwards to the right.

INSPIRED BY LIFE

9. MARKET EQUILIBRIUM AND CHANGES IN MARKET EQUILIBRIUM

In this section, we will discuss market equilibrium and changes in market equilibrium,

Meaning of equilibrium

The word equilibrium is derived from the Latin word "aequilibrium" which means equal balance. This means that opposite forces and tendencies will neutralize with each other when in the state of even balance. The position in rest will always be characterized by absence of change. The economic plans will be in complete agreement when it is in the state of even balance at various market participants so that no one has the tendency to revise or alter decisions. In the words of professor Mehta: "Equilibrium denotes in economics absence of change in movement."

Market equilibrium

There are two kinds of approaches in market equilibrium. They are as follows:

1. Partial equilibrium approach
2. General equilibrium approach

The partial equilibrium approach means the price of a single commodity will be determined keeping the prices of another commodity constant.

The general equilibrium approach means the price determination of all the goods and factors will be mutual and simultaneous. Thus, it explains a multi-market equilibrium position.

Even before Marshall, there were some disputes among economists on whether the force of demand or the force of supply is more important in determining price. Marshall has given equal importance to both demand and supply in the determination of value or price. He compared supply and demand to a pair of scissors – "We might as reasonably dispute whether it is the upper or the under blade of a pair of scissors that cuts a piece of paper, as whether value is governed by utility or cost of production." Thus, neither the upper blade nor the lower blade taken separately can cut the paper; both have their importance in the process of cutting. Likewise, neither supply alone nor demand alone can determine the price of a commodity; both are equally important in the determination of price. But the relative

importance of the two may vary depending upon the time under consideration. Thus, the demand of all consumers and the supply of all firms together determine the price of a commodity in the market.

Equilibrium between demand and supply price

Equilibrium between demand and supply price is obtained by the interaction of these two forces. Price is an independent variable. Demand and supply are dependent variables. They depend on price. Demand varies inversely with price, a rise in price causes a fall in demand and a fall in price causes a rise in demand. Thus, the demand curve will have a downward slope indicating the expansion of demand with a fall in price and contraction of demand with a rise in price. On the other hand, supply varies directly with the changes in price, a rise in price causes a rise in supply and a fall in price causes a fall in supply. Thus, the supply curve will have an upward slope. At a point where these two curves intersect with each other, the equilibrium price is established. Therefore, 'Equilibrium Price', is a price at which the quantity demanded of a given commodity equals the quantity supplied. This can be explained with the help of a table and a diagram.

Table 4.3 shows the impact of demand and supply on the price of a product. Figure 4.11 depicts the interdependence between the demand curve and the supply curve.

Table 4.3: Demand, Supply and Price Change

Price inRs	Demand in Units	Supply in Units	State Of Market	Pressure on price
30	5	25	Excess Supply	P↓
25	10	20	Excess Supply	P↓
20	15	15	Equilibrium	Neutral
10	20	10	Excess Demand	P↓
5	30	5	Excess Demand	P↓

In table 4.3, at Rs. 20, the quantity demanded is equal to the quantity supplied. As this price is agreeable to both the buyers and the sellers, there will be no tendency for it to change; this is called the equilibrium price. Let us understand why no other price can be the equilibrium price.

Suppose the price falls to Rs.5, the buyers will demand 30 units while the sellers will supply only 5 units. Excess demand over supply pushes the price upwards until it reaches the equilibrium position where supply is equal to demand.

On the other hand, if the price rises to Rs. 30 the buyers will demand only 5 units while the sellers are ready to supply 25 units. Sellers compete with each other to sell more units of the commodity. Excess supply over demand pushes the price downwards until it reaches equilibrium. This process will continue till the equilibrium price of Rs. 20 is reached.

Thus, the interactions of supply and demand forces acting upon each other restore the equilibrium position in the market.

In the diagram DD is the demand curve, SS is the supply curve. Demand and supply are in equilibrium at point E where the two curves intersect each other. OQ is the equilibrium output. OP is the equilibrium price.

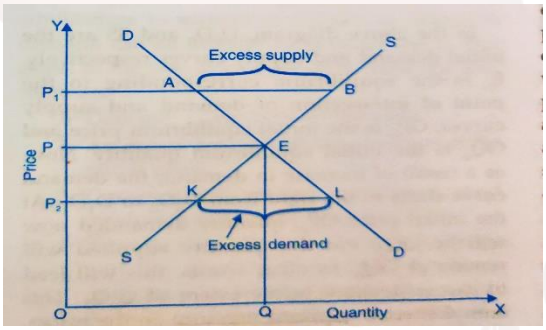


Figure 4.11: Inter-dependence of Demand Curve and Supply Curve

Source: Economics by DK Sethi

When Price Rises,

If the price rises to OP1, the quantity demanded decreases from PE or OQ to P1A. Now the consumers who could afford the commodity at OP price will not be able to purchase it at a higher price. On the other hand, supply will increase to P1B as sellers would like to supply more at a higher price to earn more profits. This increase in quantity supplied and decrease in quantity demanded will create a situation of Excess Supply AB.

The excess supply forces the sellers to reduce the price to attract more buyers so

As to dispose the excess stock. Thus, as a result of competition among sellers, the price eventually becomes equal to the equilibrium price.

When the price decreases,

If the price falls to OP2, the quantity demanded will increase to P2L.

Many consumers who were not able to afford the commodity at the higher price OP1 will now purchase it at OP2, a lower price.

The quantity supplied on the other hand would fall from PE or OQ to P2K.

Demand being more than supply, there emerges an excess demand equal to KL

This excess demand will push the price up towards the equilibrium price OP

As the excess demand means the consumers will not get the commodity at price OP2.

In an attempt to get this commodity, the consumers are willing to pay a higher price.

Therefore, as a result of the competition among buyers the price rises and eventually reaches the equilibrium price.

Quick Recap - When price rises

At price OP and Quantity OQ is the Point of Price Equilibrium i.e., 'E'.

If the Price increases to OP1 Quantity supplied increases to P1B.

Quantity demanded falls to P1A. Creates AB excess Supply.

To sell the excess supply of AB sellers reduce the price to OP. We attain equilibrium at OP price where quantity supplied and Price is the same.

Quick Recap - When price falls

At price OP and Quantity OQ is the Point of Price Equilibrium i.e., 'E'.

If the price decreases to OP2 Quantity demanded increases to P2L.

Quantity supplied falls to P2K. Creates KL an excess demand.

Here customers are willing to pay a higher price OP to buy the commodity

Due to KL excess demand Price rises to OP We attain equilibrium at OP price where the quantity demanded and Price is the same.

Changes in market equilibrium

The changes in equilibrium price will occur when there will be shift either in demand curve or in supply curve or in both.

a. Effects of shift in demand

Demand changes when there is a change in the determinants of demand like the income, tastes, prices of substitutes and complements, size of the population, etc. If demand rises due to a change in any one of these conditions, the demand curve shifts upward to the right. If, on the other hand, demand falls, the demand curve shifts downward to the left. Such a rise and fall in demand are referred to as increase and decrease in demand.

Every time there is a shift in the demand curve, a new equilibrium point is created as it intersects with the supply curve at a different level. Figure 4.12 depicts the effects of shift in demand on market equilibrium.

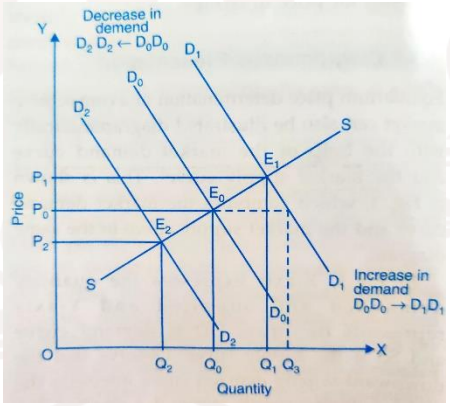


Fig 4.12 Effects of shift in demand on market equilibrium.

Source: DK Sethi

D0D0 is the demand curve and SS is the supply curve. E0 is the Equilibrium or point of intersection of demand and supply curve at OPo price and OQo quantity.

When demand increases the Demand curve shifts from D0D0 to D1D1.

The demand curve shifts and at the initial price OPo quantity now demanded will be OQ3 but the quantity supplied will be OQ0 because only the demand curve has shifted resulting in a change in demand whereas quantity supplied remains the same.

The excess demand created is Q0Q3 which will create an upward pressure on the prices. The price thus continues to rise till it reaches OP1 where the demand and supply will be equal when D1D1 intersects at SS with a new Equilibrium point at E1

An Increase in the demand for a commodity, I.e., a rightward shift in the demand curve, causes an increase in both equilibrium price and equilibrium quantity bought and sold

On the other hand, when there is a fall in the demand or leftward shift of the demand curve, both equilibrium price and equilibrium quantity will fall

b. Effects of Shift in Supply

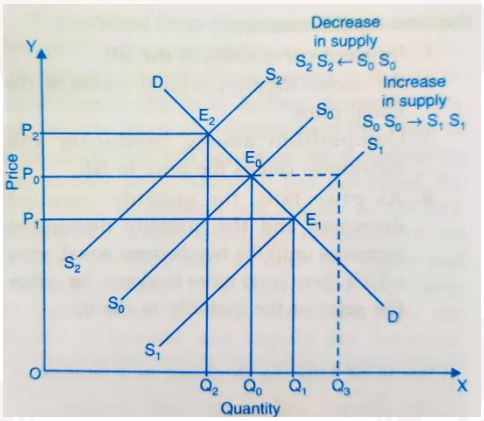


Fig 4.13 Effects of shift in Supply

Source DK Sethi Economics

DD is the demand curve and S0S0 is the supply curve.

The point of intersection of both curves is at E0 with Price OPo and quantity OQo.

If the supply increases due to better production techniques. The supply curve shifts to the right from Soso to S1S1.

The shift indicates that a larger quantity is supplied at each possible price. Now at price OPo the quantity demanded remains unchanged at OQo, but the supply increases to OQ3.

This creates an excess supply leading to a fall in price.

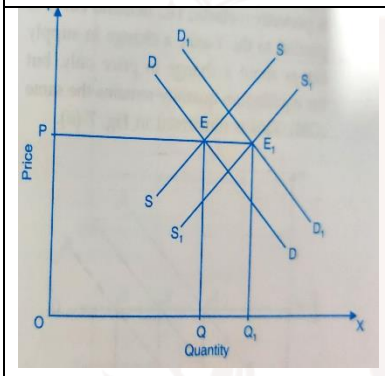
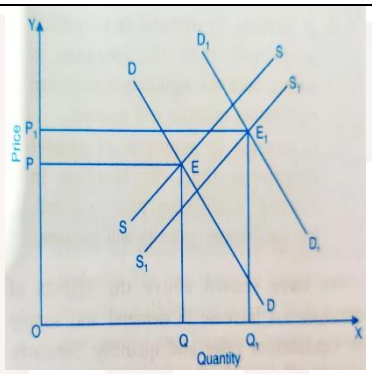
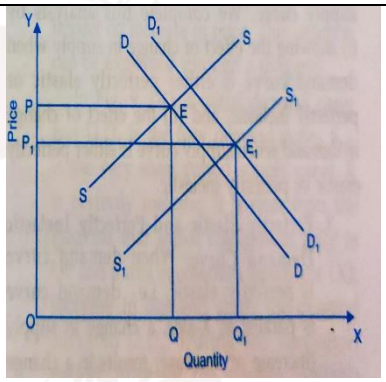
In this way a new equilibrium would emerge at E1 corresponding to the intersection of DD curve and S1S1 supply curve where OP1 is the new equilibrium price and OQ1 is the new equilibrium quantity.

An increase in supply or a rightward shift of the supply curve of a commodity causes a decrease in the equilibrium price and an increase in the equilibrium quantity bought and sold.

A decrease in supply will result in a leftward shift of the supply curve. Equilibrium price will rise and Equilibrium quantity will fall

c. Effects of changes in both demand and supply

Table showing the three scenarios of changes in both demand and supply

Equilibrium price remains the same	Equilibrium price rises	Equilibrium price falls
		
An increase in demand shifts the curve from DD to D1D1 and Supply also increases to S1S1. Equilibrium quantity increases from OQ to OQ1 but price remains the same at OP	When demand increases by a larger proportion than supply. Equilibrium shifts from E to E1. Now at E1 D1D intersects with S1S1. Hence price increases to Op1 and Quantity increases to OQ1	When the relative increase in demand is smaller than increase in supply, the equilibrium price falls. Equilibrium shifts from E to E1. Equilibrium price falls from OP to OP1 but equilibrium quantity increases from OQ to OQ1
When demand and supply increase equally equilibrium price remains unchanged but the equilibrium quantity increases	When increases in demand is more than increase in supply, Equilibrium Price and Equilibrium Quantity both increases.	When increases in supply is more than increase in demand, price tends to fall but the equilibrium quantity will increase.

Source: Table and Graphs from DK Sethi Economics

10. CONSUMER SURPLUS AND PRODUCER SURPLUS

In this section, we will discuss the consumer surplus and producer surplus. In a monetary economy, we measure the utility of a commodity with the help of price. The price we pay for a commodity basically depends on its worthiness and utility. If a product possesses higher utility, then it would command a higher price and vice versa. The law of Equi-marginal utility states that the price paid for a commodity should be equal to its marginal utility. This is one of the basic conditions for consumer's equilibrium. At the point of equilibrium, there will neither be higher utility nor lower utility, but

Marginal Utility = Price.

In real life, a consumer may not always act according to the law of Equi-marginal utility. The balance between the price and utility may not be maintained. Sometimes, the consumer may get lower utility from a commodity when compared to the price being paid for it [M.U. < price]. In some other cases, the consumer may get a higher utility [M.U. > price].

When the satisfaction obtained by the consumer is much more than the price being paid for the commodity, the consumer will be enjoying a surplus. The excess or surplus satisfaction enjoyed by a consumer over and above the price being paid for the product is technically described as consumer's surplus in economics. It is essentially found in the purchase of useful and very cheap articles like salt, postcard, newspaper, matchbox, and many other durable and non-durable articles, etc. In all these cases, we receive more utility than we pay for them.

Consumer's surplus may be defined as the excess of what a consumer is willing to pay over what he actually does pay (rather than go without the good). It is the excess of price which a person would be willing to pay over what he actually does pay. Consumer surplus is the economic measure of surplus satisfaction.

The concept may be explained with the help of a formula:

Consumers' surplus = What we are prepared to pay – [minus] What we actually pay

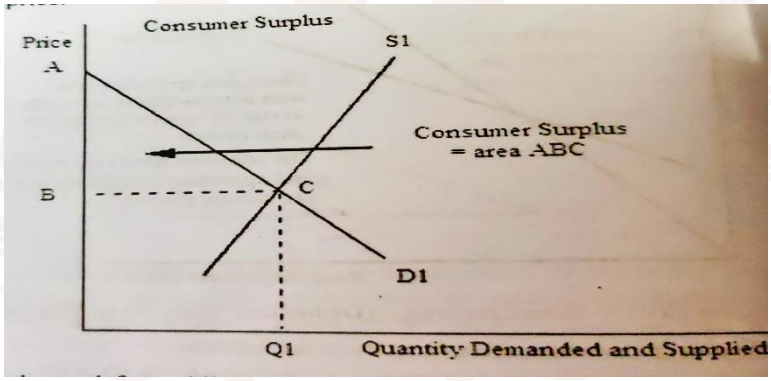
It is the difference between ex-ante and ex-post satisfaction. It is also the difference between the potential price and actual price. Hence, it is the difference between the value of the total utility that

may be derived from the consumption of a product and the total amount of money spent on that product.

It is essential to note that there is an inverse relationship between price and consumer surplus. If price rises, consumer's surplus falls and vice versa.

It is a product of opportunities and environment. For example, in developed countries consumers' surplus is high, because with a small amount of money the consumers can buy larger quantities of goods and services. On the contrary, in developing countries, consumers' surplus is low because with more amount of money, the consumers can purchase little amount of goods and services.

The concept of consumer surplus helps in making various types of cost- benefit analysis of public investments, which determines the amount of public welfare. This concept is usefully employed in most branches of economic activities.



Graph on Consumer Surplus,

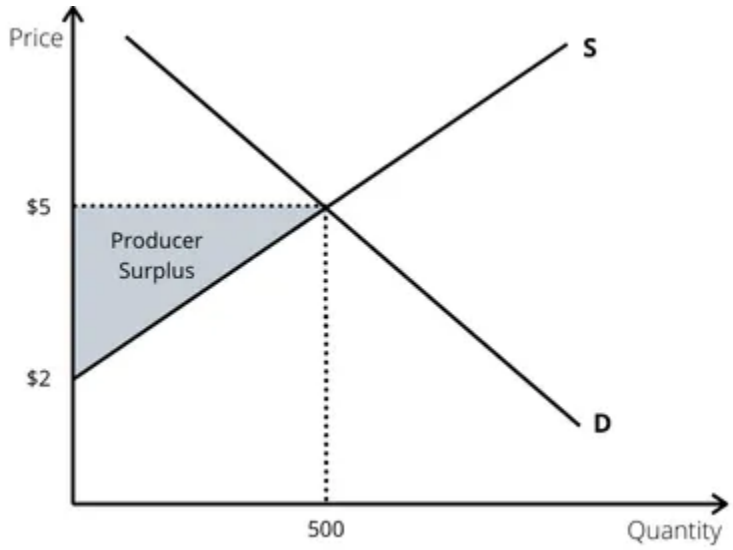
Source: Economics for managers by Geetha M Rajaram

Graph on Consumer Surplus above shows that the price expected to pay by the consumer is OA for OQ1 quantity but when the consumer only pays OB price for OQ1 quantity this is when the consumer surplus is observed as the area ABC on the graph. The satisfaction obtained by the consumer which is much more than the price being paid for the commodity, the consumer will be enjoying a surplus.

Producer Surplus:

Producer surplus is a parallel concept to consumer surplus. Producer's surplus is owner's surplus. It may be defined as the excess or surplus income received by a seller on the price at which he is willing to sell a product. It is the difference between the actual price at which he is selling and the price at

which he is willing to sell. Hence, it arises when the actual price received exceeds the minimum price that the seller is ready to accept.



Graph on Producer Surplus,

Source: Economics for managers by Geetha M Rajaram

As per the graph on Producers surplus, the producer expects an output of 500 units at a price fixed at $2 which is the minimum price but if the producer manages to sell 500units at $5 then the producer gets an additional income of $3 I.e., $5-$2 = $3 and this is the producer surplus

The producer surplus is the difference between what the producer sells its goods for and the minimum price it would be willing to sell for. In other words, because the producer is selling at a higher price than they would accept, a 'producer surplus' is created. The surplus itself is the difference between the two values.

The producer's surplus is equal to the value that the seller is actually receiving minus the value that the seller is ready to receive. It is the difference between ex-post and ex-ante income received. Depending on market situations, producers try to convert consumers' surplus into producers' surplus and consumers would try to convert producers' surplus into consumers' surplus.

SELF ASSESSMENTS QUESTIONS – 2

Fill in the blanks:	
8	The absence of change in movement in economics is called _____ .
9	The quantity bought and sold at equilibrium is called _____ .
10	There will be change in _____ when there is a shift in either demand curve or supply curve or both.
11	If the price increases there will be _____ in consumer surplus.
12	_____ is used to judge the desirability of public investment of any public projects or investment.

True or False Questions	
vi)	With a given supply curve, a decrease in demand leads to a decrease in equilibrium price and a decrease in equilibrium quantity.
vii)	Increase in supply will always lead to a decrease in equilibrium price.
viii)	If the actual market price is greater than the equilibrium price, there would be a downward pressure on price.
ix)	Equilibrium price and quantity can be calculated by equating the demand-price and supply-price functions.
x)	Equilibrium price is determined by the Government.

Activity - 1

From any nearby store, select ten products and identify which determinants of supply have affected a change in the quantity supplied to the store during the last couple of months.
Hint: Use the theoretical concept and work on it.

11. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- The management of a business firm should have a clear understanding of the supply and demand conditions in the market to have effective control of the business.
- Supply refers to the quantity of a commodity offered for sale at a particular price during a given period of time. Supply is different from production and stock. A supply schedule is a tabular representation of different quantities of a commodity supplied at varying prices and the supply curve is drawn on the basis of supply schedule which shows the direct relationship that exists between price and supply.
- Thus, the supply curve will have a positive slope. The law of supply states that 'other things being constant', a rise in price causes extension of supply and a fall in price causes contraction of supply. There are a few exceptions to this law. There will be a shift or a change in supply when the determinants of supply like the natural factors, techniques of production, cost of production, government policy, monopoly power, prices of related goods, number of sellers, etc., change.
- In order to regulate production and supply efficiently, management should have proper knowledge of the concept of elasticity of supply. Elasticity of supply refers to the responsiveness of supply to a change in price. It is influenced by a number of factors like the period of time under consideration, availability and mobility of factors of production, technological improvements, cost of production, number of sellers, prices of related goods, etc.
- Modern economics is sometimes called equilibrium analysis. The market is in equilibrium when there is a balance between supply and demand forces. The price and output determined by the interaction of supply and demand forces are called the equilibrium price and the equilibrium output. There will be a change in the equilibrium price and output when there is a change in demand or change in supply or change in both demand and supply.
- Understanding the position of market equilibrium is of very great importance in the decision-making process of the firm.

12. GLOSSARY

Consumer surplus	-	Excess of what a consumer is willing to pay over what he actually does pay.
Elasticity of demand	-	Sensitiveness or responsiveness of supply to a given change in price.
Law of supply	-	Other things remaining constant, the quantity supplied varies directly with the price..
Stock	-	Various amounts which are offered for sale at a particular price during a given period of time.
Supply curve	-	Geometrical representation of the supply schedule.
Supply schedule		Tabular representation of different quantities of a commodity supplied at varying prices.

INSPIRED BY LIFE

13. TERMINAL QUESTIONS

1. State the law of supply.
2. Explain the determinants of supply.
3. What is elasticity of supply?
4. Explain the factors determining elasticity of supply.
5. Define the term equilibrium.

14. ANSWERS

Self-Assessment Questions

1. Potential supply
2. Upwards
3. Law of supply
4. Price
5. Right
6. Inelastic
7. Supply schedule
8. Equilibrium
9. Equilibrium quantity
10. Equilibrium Price
11. Decrease
12. Cost Benefit Analysis

True or False

i) False
ii) False
iii) True
iv) False
v) False
vi) True
vii) False
viii) True
ix) True
x) False

Terminal Questions

1. A rise in price tends to increase supply and a fall in price tends to reduce it. There is a functional relationship between supply and price. Mathematically S = F (P). Refer to Section 2.
2. There are many determinants that bring about changes in supply. Among them, the important factors are: natural factors, change in techniques of production, cost of production, prices of related goods Refer to Section 2.
3. It refers to the sensitivity or responsiveness of supply to a given change in price. In short, it measures the degree of adjustability of supply to a given change in price of a product. Refer to section 5.
4. The factors determining the elasticity of supply are as follows: Time period, Availability and mobility of factors of production, Technological improvements, Cost of production. Refer to Section 6.

15. CASE STUDY – ECONOMICS IN ACTION

Persistent Food Inflation Warrants Monetary Policy Responses

RBI

PUNE: Deputy Governor of Reserve Bank of India, Subir Gokarn said that the drivers of food inflation have changed from the pair of sugar and cereals during the last four decades of the previous century to proteins, fruits and vegetables during last four years. He also asserted that the situation of persistent high food inflation warrants monetary policy responses to keep the overall inflation in check.

Mr. Gokarn said that, "The more worrying attribute of food inflation is persistence. In the four decades in 1960's, the drivers of food inflation were cereals and sugar with a supporting role being played by proteins and fruits and vegetables in the second half of the period". Mr. Gokarn added "While contribution from proteins has gone down somewhat in 2011-12, which is the basis of the declines we have seen in food inflation in the most recent date releases, the pressure from fruits and vegetables have sustained."

Mr. Gokarn said on the demand side that people eat better as they get richer is a simple assertion. But on the supply side, we have problems. The productivity of pulses has oscillated around 600 kg/hectare for the last decade and a half. "There are several states, in which pulses may constitute an important part of the typical diet, whose productivity is significantly below the national average. If overall productivity is to be improved, a strategy which focuses on the specific bottlenecks in these states is probably the best way to go about it" he said.

Mr. Gokarn pointed out the rising wages of agricultural labour and the stocks of agricultural commodities as two important factors exerting influence on their prices. "Wages rising faster than productivity can only result in rising prices, if producers cannot substitute other inputs for labour. There is an important relationship between price dynamics and stocks. Periods in which stocks have been high show relatively low rates of price increase," he said.

Lastly, he pointed out the declining trend in the long period average (LPA) of monsoon rainfall. "This means that normal monsoons are delivering less water than in the past". He said that the implication of persistent supply pressures on the economy are not very good for maintaining balance between

fast growth and low inflation. "While transitory episodes of food inflation do not warrant a monetary policy response, there are strong justifications for acting in the face of more persistent ones, if the objective is to keep overall inflation in check".

Discussion Questions:

1. What are the factors influencing the demand and supply of foods such as fruits, vegetables and pulses?
2. What is the nature of the impact of lower supply and stocks of food-on-food prices?
3. What is the nature of the impact of higher wages on food prices?

(***Source***: *Economic Times, December 9, 2011*)

Hint: With the help of the theoretical concepts build your views in this case study.

16. REFERENCES

- McConnell, Campbell R. (2005), Economics: Principles, Problems, and Policies, Edition16, McGraw-Hill/Irwin.

E-References:

- www.economictimes – retrieved on- December 9, 2011